

Email Header Analyzer

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Course: BCA – Second Year

Project Title: Email Header Analyzer

Technology Used: Python (Flask), HTML, CSS, Bootstrap

1. Introduction

The Email Header Analyzer is a web-based tool designed to identify and analyze important security information within email headers. This project helps detect phishing, spoofing, or suspicious emails by verifying sender information and authentication records such as SPF, DKIM, and DMARC.

2. Objective

- To analyze email headers and extract key security details.
- To identify potential risks based on mismatched or failed validations.
- To display results in a simple and understandable web interface.
- To promote awareness about safe email practices.

3. Technologies Used

Frontend: HTML, CSS, Bootstrap

Backend: Python Flask

Tools: Visual Studio Code, GitHub

File Handling: Text (.txt) and Email (.eml) formats

4. System Requirements

Software: Python 3.10 or higher, Flask framework, Browser

Hardware: Minimum 2GB RAM, Intel i3 or higher

5. Methodology

1. User pastes or uploads an email header.
2. Flask reads the header and extracts fields (From, Return-Path, SPF, DKIM, DMARC, Subject, IP).
3. System checks for mismatches, failed validations, and suspicious subjects.
4. Risk score displayed as Low, Medium, or High Risk.

6. Results

The analyzer successfully identifies suspicious email headers and shows risk results in a colorful and structured interface.

7. Conclusion

The Email Header Analyzer provides a practical way to understand email security basics. It helps users identify fake or spoofed emails and promotes cybersecurity awareness.

8. Future Enhancements

- Add IP geolocation tracking.
- Display mail server hop timeline.
- Integrate machine learning for smarter classification.
- Generate downloadable PDF reports.

9. References

Flask Documentation: <https://flask.palletsprojects.com>

Python Docs: <https://docs.python.org>

Email Header Format (RFC 5322)